

## **KENYA INSTITUTE OF PRIMATE RESEARCH**

Ministry of Health

# **Institutional Data Governance Development Plan**

2026–2027

*This Development Plan provides the strategic road-map for the progressive development and implementation of the KIPRE IDGF in alignment with the KIPRE Strategic Plan (2023–2027).*

**Document control**

| Document information           |                                                          |
|--------------------------------|----------------------------------------------------------|
| <b>Document title</b>          | Institutional Data Governance Framework Development Plan |
| <b>Document number</b>         | KIPRE/DSAS/DGF/PLN/001                                   |
| <b>Document owner</b>          | Data Science and Analytics Section (DSAS)                |
| <b>Prepared by</b>             | Patrick W. Mwaura                                        |
| <b>Reviewed by</b>             | Director, Research and Product Development               |
| Signature and Date             | _____                                                    |
| <b>Approved by</b>             | Director General                                         |
| Signature and Date             | _____                                                    |
| <b>Effective date</b>          | .....                                                    |
| <b>Next review date</b>        | .....                                                    |
| <b>Distribution</b>            | Internal                                                 |
| <b>Document classification</b> | Internal Use Only                                        |

**Revision history**

| Version | Date         | Description of revision | Responsible officer |
|---------|--------------|-------------------------|---------------------|
| 1.0     | 31 July 2026 | Initial issue.          | Patrick W. Mwaura   |
|         |              |                         |                     |
|         |              |                         |                     |
|         |              |                         |                     |

## Executive summary

This Development Plan provides the implementation road-map for establishing the KIPRE IDGF. Building upon the approved Data Management and Protection Policy. The Plan presents a structured approach for progressively developing the governance policies, operational procedures, standards, guidelines and supporting implementation documents required to establish a comprehensive institutional data governance program.

Implementation of the IDGF will be undertaken through three progressive phases.

**Phase I** focuses on establishing institutional data protection compliance by operationalizing the approved Policy through the development of an implementation strategy, governance structures, a network of Data Protection Champions, priority SOPs and supporting implementation documents.

**Phase II** expands the governance framework through the development of additional governance policies and supporting governance instruments, while

**Phase III** institutionalizes the completed framework through organization-wide implementation, capacity building, monitoring, evaluation and continual improvement.

Successful implementation of this Development Plan will establish a coordinated, sustainable and enterprise-wide institutional data governance program that strengthens regulatory compliance, improves institutional accountability, enhances the quality and stewardship of institutional data, and supports evidence-based decision-making, responsible research and innovation in advancing KIPRE's strategic mandate.

## Contents

|                                           |          |
|-------------------------------------------|----------|
| <b>List of abbreviations</b>              | <b>5</b> |
| <b>1 Background context</b>               | <b>6</b> |
| <b>2 Introduction to the Plan</b>         | <b>8</b> |
| 2.1 Plan purpose and objectives . . . . . | 8        |
| 2.2 Plan Structure . . . . .              | 9        |

## PART I

|                                                                       |           |
|-----------------------------------------------------------------------|-----------|
| <b>Status of institutional data governance landscape</b>              | <b>10</b> |
| 2.3 Current state of institutional data governance . . . . .          | 10        |
| 2.4 Existing institutional governance structures . . . . .            | 11        |
| 2.4.1 Board of Directors . . . . .                                    | 11        |
| 2.4.2 Director General . . . . .                                      | 11        |
| 2.4.3 ISREC . . . . .                                                 | 11        |
| 2.4.4 Directorate of Research and Product Development(RPDD) . . . . . | 11        |
| 2.4.5 DSAS . . . . .                                                  | 12        |
| 2.4.6 DPO Office . . . . .                                            | 12        |
| 2.4.7 ICT . . . . .                                                   | 12        |
| 2.4.8 Corporate support Divisions . . . . .                           | 12        |
| 2.4.9 Collaborating institutions . . . . .                            | 12        |
| 2.5 Proposed institutional data governance architecture . . . . .     | 13        |
| 2.6 Proposed institutional governance document suite . . . . .        | 13        |
| 2.6.1 Governance policies . . . . .                                   | 16        |

## PART II

|                                                                                    |           |
|------------------------------------------------------------------------------------|-----------|
| <b>Development road-map</b>                                                        | <b>18</b> |
| 2.7 Development approach . . . . .                                                 | 18        |
| 2.8 Guiding principles . . . . .                                                   | 18        |
| 2.9 Governance maturity pathway . . . . .                                          | 20        |
| 2.10 Critical success factors . . . . .                                            | 20        |
| 2.11 Phased development strategy . . . . .                                         | 20        |
| 2.12 Phase I: Operationalizing the Data Management and Protection Policy . . . . . | 22        |
| 2.12.1 Objectives . . . . .                                                        | 22        |
| 2.12.2 Data protection compliance implementation strategy . . . . .                | 22        |
| 2.12.3 SOPs inventory . . . . .                                                    | 23        |
| 2.12.4 Priority SOPs for the DPO Office . . . . .                                  | 26        |
| 2.12.5 Supporting implementation documents . . . . .                               | 26        |
| 2.12.6 Expected outputs . . . . .                                                  | 28        |
| 2.13 Phase II: Expansion of the governance framework . . . . .                     | 29        |
| 2.13.1 Purpose of Phase II . . . . .                                               | 29        |

|                                                            |           |
|------------------------------------------------------------|-----------|
| 2.13.2 Governance policies to be developed . . . . .       | 29        |
| 2.13.3 Supporting governance instruments . . . . .         | 30        |
| 2.13.4 Governance structures and stewardship . . . . .     | 30        |
| 2.14 Phase III: Institutionalization of the IDGF . . . . . | 31        |
| 2.14.1 Framework integration . . . . .                     | 31        |
| 2.14.2 Capacity building . . . . .                         | 31        |
| 2.14.3 Monitoring and evaluation . . . . .                 | 31        |
| 2.14.4 Continuous improvement . . . . .                    | 31        |
| <b>3 Overall implementation road-map</b>                   | <b>32</b> |
| <b>4 Conclusion</b>                                        | <b>32</b> |
| <b>5 References</b>                                        | <b>33</b> |
| <b>References</b>                                          | <b>33</b> |

## List of Tables

|                                                                                                                |    |
|----------------------------------------------------------------------------------------------------------------|----|
| 1 Institutional data governance document hierarchy. . . . .                                                    | 15 |
| 2 Governance policy portfolio proposed under the Institutional Data Governance Framework. . . . .              | 16 |
| 3 Proposed SOPs required to operationalize the Data Management and Protection Policy. . . . .                  | 25 |
| 4 Priority SOPs for the DPO Office. . . . .                                                                    | 26 |
| 5 Proposed supporting implementation documents for the Phase I priority Standard Operating Procedures. . . . . | 28 |
| 6 Proposed governance policies for development under Phase II. . . . .                                         | 29 |
| 7 Proposed governance structure and stewardship responsibilities under the IDGF. . . . .                       | 30 |
| 8 Overall implementation road-map for the IDGF. . . . .                                                        | 32 |

## List of Figures

|                                                                                                                                                                                                                        |    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 1 Proposed architecture of the Institutional Data Governance Framework showing the governance hierarchy from institutional governance structures through governance instruments and implementation mechanisms. . . . . | 14 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|

**List of abbreviations**

|       |                                                        |
|-------|--------------------------------------------------------|
| DPA   | Data Protection Act, 2019                              |
| GDPR  | Data Protection (General) Regulations, 2021            |
| AI    | Artificial Intelligence                                |
| DGF   | Data Governance Framework                              |
| DMP   | Data Management Plan                                   |
| DPO   | Data Protection Officer                                |
| DSAS  | Data Science and Analytics Section                     |
| ICT   | Information and Communication Technology               |
| IDGF  | Institutional DGF                                      |
| ISREC | Institutional Scientific and Research Ethics Committee |
| KIPRE | Kenya Institute of Primate Research                    |
| ODPC  | Office of the Data Protection Commissioner             |
| PI    | Principal Investigator                                 |
| RDM   | Research Data Management                               |
| SOP   | Standard Operating Procedure                           |

## 1 Background context

Data has become one of the most valuable strategic assets within modern organizations. Across the research enterprise, institutional data support scientific discovery, innovation, operational management, regulatory compliance and evidence-informed decision-making.

Advances in artificial intelligence (AI), cloud computing, high-performance computing, advanced analytics and interconnected digital ecosystems have significantly increased both the volume and complexity of data generated by research institutions. Consequently, the ability to govern institutional data effectively has become a strategic institutional capability that directly influences research quality, operational efficiency, accountability and long-term sustainability.

Globally, data governance has evolved beyond traditional Information Technology (IT) management to become an enterprise-wide governance function integrating data quality, stewardship, ownership, privacy, information security, research data management (RDM), records management, regulatory compliance and responsible innovation.

Institutions with mature data governance frameworks are better positioned to maximize the value of their data assets, strengthen institutional resilience, facilitate collaboration and respond effectively to rapidly evolving technological and regulatory environments.

The rapid advancement of AI has further elevated the strategic importance of institutional data governance. While recent developments have been characterized by increasingly capable foundation models, future AI adoption is expected to become progressively domain-specific, with organizations developing specialized AI solutions supported by high-quality institutional datasets.

For biomedical research institutions, research data are no longer merely scientific outputs but strategic institutional assets that underpin advanced analytics, translational research, precision medicine, predictive modelling and future AI-enabled scientific discovery.

Robust institutional data governance has therefore become a prerequisite for ensuring that these data remain trustworthy, interoperable, secure and suitable for responsible AI innovation.

In Kenya, the Government continues to advance an ambitious digital transformation agenda through initiatives such as the Digital Superhighway Program, expansion of e-Government services, establishment of the National Data Centre, development of Konza Technopolis and implementation of the National Artificial Intelligence Strategy. These national initiatives are supported by an evolving legal, regulatory and policy environment that places increasing emphasis on the responsible governance, protection and utilization of institutional data assets.

The governance of institutional data is anchored within several national legal and regulatory instruments, including the Constitution of Kenya, 2010 (Article 31 (Right to Privacy) and Article 35 (Access to Information)), the Data Protection Act, 2019, the Computer Misuse and Cybercrimes Act, 2018, the Access to Information Act, 2016, the Kenya Information and Communications Act, the Public Archives and Documentation Service Act, and other sector-specific legal and regulatory frameworks governing public sector information management, cybersecurity, records management and research.

Collectively, these instruments require public institutions to establish governance mechanisms capable of ensuring that institutional data are managed lawfully, securely, ethically and responsibly

throughout their life-cycle.

As Kenya's national biomedical research institution, the Kenya Institute of Primate Research (KIPRE) operates within a multidisciplinary governance environment that encompasses scientific research, public health, institutional management and regulatory compliance.

The Institute conducts research under the broader national research governance framework established by the Science, Technology and Innovation Act, 2013, while simultaneously complying with national legal and regulatory requirements governing the protection, management and utilization of institutional information and personal data.

In executing its mandate, KIPRE generates and manages diverse categories of scientific, laboratory, surveillance, administrative and corporate data. These data constitute critical institutional assets that support research, institutional operations, strategic planning, national health priorities and collaborative partnerships.

Effective governance of these data assets is therefore fundamental to the Institute's ability to fulfil its mandate while maintaining scientific integrity, regulatory compliance, institutional accountability and public trust.

Recognizing the strategic importance of institutional data, the KIPRE Strategic Plan (2023–2027) identifies the establishment of an Institutional Data Governance Framework (IDGF) as a strategic institutional initiative for strengthening the governance, management and stewardship of institutional data assets.

To better understand the Institute's existing data management environment, the Data Science and Analytics Section (DSAS) conducted an institution-wide Data Management Needs Assessment involving research, technical and administrative staff.

The assessment demonstrated widespread recognition of the strategic value of institutional data while simultaneously identifying opportunities for strengthening institutional governance through standardized policies, harmonized operating procedures, improved coordination of data management activities, enhanced data quality practices, institutional capacity development and clearly defined governance structures.

Building upon these findings, the Institute has established important governance foundations through the approval of the Data Management and Protection Policy and subsequent registration with the Office of the Data Protection Commissioner (ODPC) as both a Data Controller and a Data Processor. These milestones represent significant progress towards strengthening institutional data governance and regulatory compliance.

However, effective institutional data governance extends beyond compliance with data protection legislation. It requires an integrated governance system comprising governance policies, governance structures, Standard Operating Procedures (SOPs), standards, guidelines, implementation tools, monitoring mechanisms and institutional accountability arrangements that collectively govern institutional data throughout its life-cycle.

Establishing these governance instruments in a coordinated and systematic manner is essential for supporting responsible data stewardship, improving data quality, facilitating responsible data sharing, strengthening institutional decision-making and enabling future digital transformation initiatives.



The development of the KIPRE IDGF therefore represents the next strategic step in the Institute's governance maturity journey. This Development Plan provides the road-map for progressively establishing the Framework by defining the governance architecture, identifying the governance instruments required for implementation and presenting a phased approach through which KIPRE will institutionalize an integrated, sustainable and enterprise-wide data governance program.

## **2 Introduction to the Plan**

The IDGF is the overarching institutional governance system through which KIPRE governs the management, protection, quality, stewardship and utilization of institutional data assets. The Framework establishes the governance architecture required to coordinate governance structures, policies, standards, Standard Operating Procedures (SOPs), guidelines and supporting implementation instruments necessary for effective governance of institutional data throughout its life-cycle.

The IDGF provides an enterprise-wide governance approach that extends beyond compliance with data protection legislation by integrating broader governance domains, including research data management, data quality, information governance, data stewardship, responsible data sharing and emerging digital technologies. Together, these governance components provide the institutional mechanisms required to ensure that data remain secure, reliable, interoperable, accessible and fit for purpose in supporting research excellence, innovation, regulatory compliance and evidence-informed decision-making.

The approval of the Data Management and Protection Policy established the first governance component of the Framework and provided the institutional foundation for strengthening data governance at KIPRE. The progressive establishment of the remaining governance components now requires a structured implementation program capable of systematically expanding this foundation into a comprehensive institutional governance framework.

This Development Plan therefore serves as the implementation blueprint for progressively establishing the IDGF. It defines the governance architecture, identifies the governance instruments required for implementation, assigns institutional responsibilities and presents a phased road-map through which the Framework will be developed, operationalized and institutionalized across the Institute.

### **2.1 Plan purpose and objectives**

The purpose of this Development Plan is to provide a structured road-map for progressively establishing and institutionalizing the IDGF at KIPRE.

Specifically, the Plan seeks to:

1. operationalize the approved Data Management and Protection Policy through a coordinated institutional implementation program;
2. establish institutional governance structures, including the Office of the Data Protection Officer (DPO), Directorate Data Governance Champions and clearly defined governance responsibilities to support implementation of the Framework;
3. provide a phased approach for developing governance policies, SOPs, standards, guidelines and supporting governance instruments required under the Framework;

4. strengthen institutional capacity for responsible data governance, regulatory compliance, risk management and data stewardship; and
5. institutionalize an institution-wide data governance program that supports research excellence, innovation, digital transformation and evidence-informed decision-making.

## 2.2 Plan Structure

The Development Plan is organized into two complementary parts that collectively provide the strategic direction and implementation road-map for establishing the IDGF.

- **Part I** presents the institutional data governance landscape by describing the existing governance environment, the proposed governance architecture and the governance instruments required for the Framework.
- **Part II** presents the implementation road-map by outlining the development approach, guiding principles and phased implementation strategy through which the Framework will be progressively developed, operationalized and institutionalized across the Institute.

Together, the two parts provide a coherent framework for progressively establishing an integrated institutional data governance program that supports effective stewardship of institutional data assets.

## PART I

### Status of institutional data governance landscape

#### 2.3 Current state of institutional data governance

The KIPRE (KIPRE) has made significant progress towards strengthening institutional data governance over recent years.

This progress has been driven by increasing reliance on digital technologies to support research, administration and institutional operations. The Institute has progressively adopted several digital systems, including the Enterprise Resource Planning (ERP) system, Human Resource Information System (HRIS), biometric attendance management system, electronic procurement (e-GP), financial management systems, laboratory information systems (LIMS), and other digital platforms supporting research and corporate functions.

These systems continue to generate substantial volumes of operational and administrative data that require coordinated governance to ensure their quality, integrity, security and effective utilisation.

At the same time, the Institute has experienced continued growth in multidisciplinary research programmes involving human participants, non-human primates, laboratory investigations, genomics, surveillance, environmental health, One Health initiatives and collaborative research with national and international partners. These activities generate diverse categories of institutional data, including research datasets, laboratory records, surveillance data, biological specimen metadata, administrative records, financial information, human resource data and other institutional information assets.

The increasing volume, diversity and strategic importance of these data require governance mechanisms that promote consistency, accountability and responsible stewardship throughout the data life-cycle.

Recognizing these emerging needs, the KIPRE Strategic Plan 2023–2027 identifies the enhancement of data management, analytics and protection strategies as a strategic institutional priority and specifically provides for the development and implementation of an IDGF. This strategic commitment reflects the Institute's recognition that data are strategic institutional assets that support research excellence, operational efficiency, innovation, regulatory compliance and evidence-informed decision-making.

In support of this strategic direction, the Institute has already established several important governance foundations. These include the approval of the Data Management and Protection Policy, registration with the ODPC as both a Data Controller and a Data Processor, establishment of the DPO Office, and completion of an institution-wide Data Management Needs Assessment to better understand existing data management practices, governance challenges and future institutional requirements.

Collectively, these initiatives provide a strong foundation upon which a comprehensive IDGF can be

progressively developed. However, while these achievements establish the Institute's initial governance environment, they do not yet constitute a complete institutional governance framework.

Several governance components—including additional policies, SOPs, standards, guidelines, forms, templates, registers and governance mechanisms—remain to be developed to ensure coordinated management of institutional data assets throughout their life-cycle.

The progressive development of these governance instruments therefore represents the next stage in strengthening institutional data governance and advancing KIPRE towards a mature, sustainable and institution-wide data governance programme.

## **2.4 Existing institutional governance structures**

Institutional data governance at KIPRE is currently supported through a number of governance structures established to fulfill specific statutory, administrative, research and operational responsibilities.

Although these structures were established independently to support different institutional functions, each contributes directly or indirectly to the governance of institutional data. Collectively, they provide an important governance foundation upon which the IDGF will be progressively developed.

### **2.4.1 Board of Directors**

The Board of Directors provides the highest level of institutional oversight and is ultimately responsible for ensuring that the Institute establishes appropriate governance, risk management and accountability mechanisms. The Board approves institutional policies and provides strategic direction for the governance of institutional resources, including information and data assets.

### **2.4.2 Director General**

The Director General is responsible for the overall implementation of institutional policies and provides executive oversight for institutional governance. The Office of the Director General is responsible for ensuring that appropriate governance structures, resources and institutional arrangements are established to support effective management of institutional data.

### **2.4.3 ISREC**

The ISREC provides oversight for the scientific and ethical conduct of research involving human participants, animals and other regulated research activities. The Committee reviews research protocols to ensure compliance with ethical principles, confidentiality requirements and applicable regulatory obligations. As research increasingly becomes data-intensive, the Committee provides an important governance interface between research ethics and institutional data governance.

### **2.4.4 Directorate of Research and Product Development(RPDD)**

The RPDD coordinates institutional research programs and oversees research implementation, monitoring and reporting. The Directorate plays an important role in ensuring that research data are generated, managed and utilized in accordance with approved institutional policies and research governance requirements.

### **2.4.5 DSAS**

The DSAS serves as the Institute's technical focal point for data management, statistical analysis, data analytics, data integration, data visualization and institutional data systems. In accordance with the KIPRE Strategic Plan 2023–2027, the Section is responsible for leading the development of the IDGF and supporting the implementation of institutional strategies relating to data management, analytics and protection.

### **2.4.6 DPO Office**

The DPO Office provides institutional oversight for compliance with the DPA and associated regulations. Following KIPRE's registration with the ODPC as both a Data Controller and a Data Processor, the Office is responsible for advising the Institute on data protection obligations, reviewing personal data processing activities, promoting institutional compliance, coordinating data protection awareness and supporting the implementation of appropriate safeguards for personal data.

### **2.4.7 ICT**

The ICT Division provides the technological infrastructure that supports institutional information systems. The Division is responsible for network administration, cybersecurity, server infrastructure, user access management, system administration, backup and recovery services, and implementation of technical safeguards necessary to ensure the confidentiality, integrity and availability of institutional information systems.

### **2.4.8 Corporate support Divisions**

Other institutional divisions, including Finance and Accounts, Human Resource and Administration, Procurement, Planning and Strategy, Internal Audit and Supply Chain Management, generate, process and maintain significant volumes of institutional data in support of their respective mandates. These divisions are responsible for ensuring that data generated within their functional areas are managed in accordance with approved institutional policies and applicable legal and regulatory requirements.

### **2.4.9 Collaborating institutions**

KIPRE collaborates extensively with national and international universities, research institutes, government agencies, development partners and funding organizations. These collaborations frequently involve the sharing, processing and management of research and institutional data, necessitating governance mechanisms that support responsible data sharing, clearly defined institutional responsibilities and compliance with applicable legal, ethical and contractual obligations.

Collectively, these governance structures demonstrate that responsibility for institutional data is distributed across multiple organizational units, each exercising governance within its respective mandate. However, institutional data frequently traverse these functional boundaries throughout their life-cycle.

The proposed IDGF is therefore intended to provide an overarching governance architecture that coordinates, harmonizes and strengthens these existing governance structures through clearly defined governance principles, roles, responsibilities and institutional governance instruments.

## 2.5 Proposed institutional data governance architecture

The IDGF is envisioned as the overarching governance architecture through which the Institute will coordinate the management, protection, quality, accessibility, sharing and utilization of institutional data assets. Rather than establishing an entirely new governance structure, the Framework will integrate existing governance mechanisms, strengthen institutional coordination and provide a coherent governance environment for managing institutional data throughout its life-cycle.

The Framework is intended to provide a common institutional approach to data governance by establishing clear governance principles, institutional responsibilities, decision-making structures and supporting governance instruments. It will facilitate harmonization of governance practices across research, laboratory, administrative and corporate functions while ensuring alignment with applicable legal, regulatory and institutional requirements.

The proposed IDGF consists of two complementary components:

- a) **Governance structures**, which provide institutional leadership, oversight, accountability and decision-making; and
- b) **Governance documents**, which provide the policies, procedures, standards and implementation tools required to operationalize governance across the Institute.

The governance structures described in the previous section will continue to provide institutional oversight within their respective mandates. The governance documents will progressively establish a structured hierarchy that translates institutional objectives into operational practice while supporting consistency, accountability and continual improvement.

The proposed governance architecture is illustrated in Figure 1.

The purpose of each governance document category is summarized in Table 1.

The governance document hierarchy provides the operational foundation upon which the IDGF will be progressively developed. Consistent with the phased implementation approach presented in this Development Plan, Phase I will focus on operationalizing the existing Data Management and Protection Policy through the development of supporting SOPs and their associated forms, templates, notices and institutional registers required for effective implementation. Subsequent phases will progressively expand the Framework through the development of additional governance policies and their corresponding implementation documents until a comprehensive, integrated and sustainable IDGF is established.

## 2.6 Proposed institutional governance document suite

The IDGF will ultimately comprise a coordinated suite of governance documents that collectively support the governance of institutional data assets throughout their life-cycle. While the Institute has already established important governance foundations, the Framework envisions the progressive development of additional governance instruments to address emerging institutional, scientific and regulatory requirements.

The proposed document suite is intended to guide future governance development and does not imply that all documents will be developed simultaneously. Rather, implementation will be undertaken in phases according to institutional priorities, available resources and strategic needs.

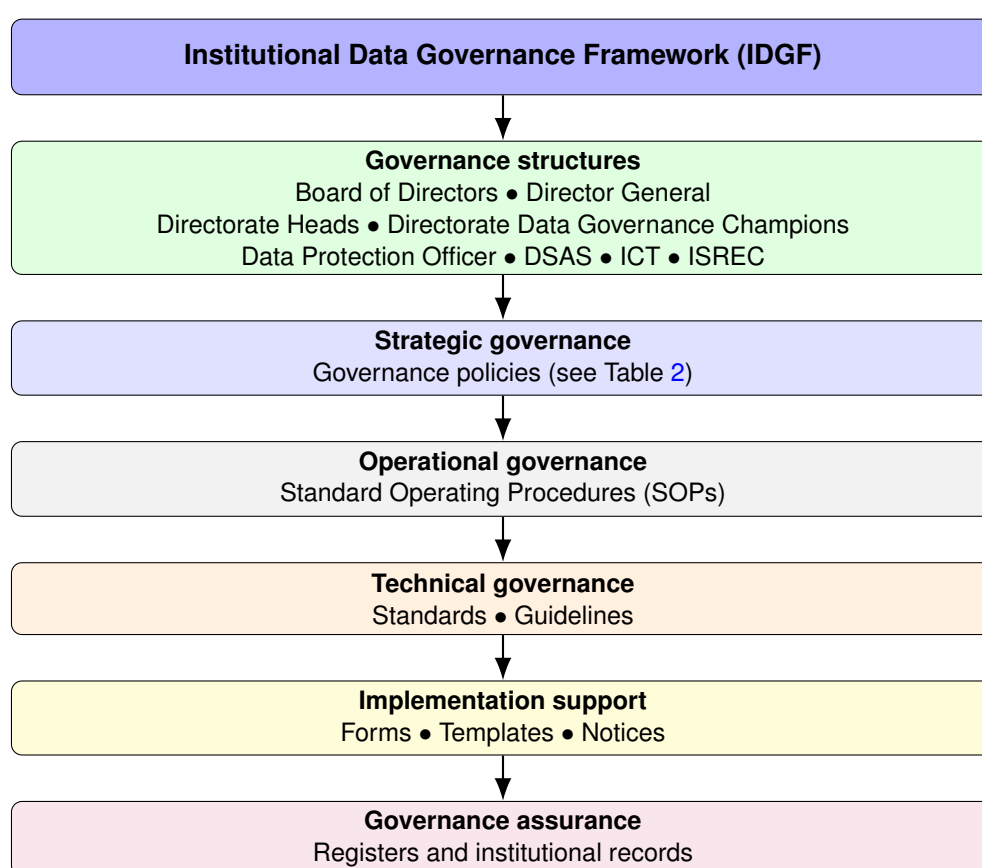


Figure 1: Proposed architecture of the Institutional Data Governance Framework showing the governance hierarchy from institutional governance structures through governance instruments and implementation mechanisms.

| Governance layer | Governance instrument                    | Purpose                                                                                                                                                                                                     |
|------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Strategic        | Policies                                 | Establish institutional governance principles, strategic direction and management commitment. Policies translate applicable legal, regulatory and institutional requirements into institutional governance. |
| Operational      | Standard Operating Procedures (SOPs)     | Define institutional workflows, responsibilities, operational controls and decision-making processes required to implement approved policies consistently across the Institute.                             |
| Technical        | Standards and guidelines                 | Provide technical specifications, minimum requirements, recommended practices and implementation guidance to promote consistency, interoperability and continual improvement.                               |
| Implementation   | Forms, templates, notices and checklists | Provide controlled instruments that standardise governance activities, support implementation and ensure consistent documentation across institutional processes.                                           |
| Assurance        | Registers and institutional records      | Provide evidence of implementation, facilitate monitoring, audit, reporting and continual improvement while demonstrating institutional accountability and regulatory compliance.                           |

Table 1: Institutional data governance document hierarchy.



## 2.6.1 Governance policies

The proposed governance policy portfolio and its legal and institutional basis are presented in Table 2.

| Governance area                    | Governance policy                         | Legal and regulatory basis                                                                                               | Impl/Dev. phase | Purpose                                                                                                                                                                       |
|------------------------------------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Privacy and data protection        | Data Management and Protection Policy     | Constitution of Kenya (Art. 31); Data Protection Act, 2019; Data Protection Regulations, 2021 (LN. 263,264,265)          | Existing        | Provides the institutional framework for the lawful collection, processing, storage, sharing, retention and protection of personal and institutional data.                    |
| Research data governance           | Research Data Management Policy           | Science, Technology and Innovation Act, 2013; NACOSTI research requirements; KIPRE Strategic Plan 2023–2027              | Phase II        | Provides governance for research data throughout the research life-cycle, including planning, stewardship, preservation, sharing and long-term archiving.                     |
| Data quality governance            | Data Quality Management Policy            | KIPRE Strategic Plan 2023–2027; ISO 8000 (series); DAMA-DMBOK2                                                           | Phase II        | Establishes institutional requirements for ensuring data quality, integrity, completeness, consistency, timeliness and fitness for use.                                       |
| Information governance             | Information Governance Policy             | Access to Information Act, 2016; Public Archives and Documentation Service Act; Kenya Information and Communications Act | Phase II        | Provides enterprise-wide governance for institutional information assets through integrated management of the information life-cycle, accountability, access and stewardship. |
| Artificial intelligence governance | Artificial Intelligence Governance Policy | Kenya National AI Strategy; Data Protection Act, 2019; UNESCO Recommendation on the Ethics of AI                         | Phase II        | Establishes principles for the responsible development, acquisition, deployment and governance of Artificial Intelligence systems.                                            |
| Open science and data sharing      | Open Science and Data Sharing Policy      | UNESCO Recommendation on Open Science; Data Protection Act, 2019; institutional and donor data-sharing requirements      | Phase II        | Promotes responsible data sharing, open science practices, collaborative research and compliance with national and international research dissemination requirements.         |

Table 2: Governance policy portfolio proposed under the Institutional Data Governance Framework.

The governance policies presented above collectively define the target policy portfolio for the proposed IDGF. While the Data Management and Protection Policy provides the Institute with an important governance foundation, the remaining policies will be progressively developed to establish a comprehensive, integrated and sustainable governance framework capable of supporting the Institute's evolving research, operational and digital transformation agenda.

The development of these policies will be undertaken in phases in accordance with institutional priorities, available resources and strategic requirements. The proposed phased development approach is presented in Part II of this Development Plan.

## PART II

### Development road-map

#### 2.7 Development approach

The development of the IDGF will be undertaken through a phased and incremental approach that recognizes the Institute's existing governance achievements while progressively expanding institutional data governance maturity.

This approach acknowledges that the establishment of a comprehensive governance framework is a continuous institutional process requiring coordinated planning, prioritization and sustained implementation rather than the development of isolated governance documents.

The proposed approach is founded on the principle of progressive governance maturity. Rather than attempting to develop all governance instruments simultaneously, the Institute will build upon its existing governance foundations by first strengthening and operationalising approved governance instruments before expanding into additional governance domains.

This strategy promotes efficient utilization of institutional resources, minimizes duplication of effort and ensures that each governance component is fully embedded before subsequent governance instruments are introduced.

The phased approach also recognizes that institutional data governance extends beyond regulatory compliance. While compliance with the DPA and related legal and regulatory requirements remains an important objective, the broader purpose of the Framework is to establish governance structures that enhance institutional decision-making, improve research quality, strengthen data stewardship, support digital transformation and maximize the strategic value of institutional data assets.

The implementation priorities presented in this Development Plan have been informed by the Institute's current governance landscape, the findings of the Institutional Data Management Needs Assessment, strategic priorities identified in the KIPRE Strategic Plan 2023–2027, and emerging national and international developments in data governance. Collectively, these provide a practical and evidence-based road-map for progressively strengthening institutional governance of data.

The development road-map is organized into three complementary phases. Phase I focuses on operationalising the existing Data Management and Protection Policy through the development of its supporting operational documentation. Phase II expands the governance portfolio through the development of additional governance policies and their supporting governance instruments. Phase III consolidates these governance instruments into a fully integrated IDGF supported by institutional capacity building, implementation, monitoring and continual improvement.

#### 2.8 Guiding principles

The development and implementation of the IDGF shall be guided by principles that promote responsible, consistent and sustainable governance of institutional data assets. These principles provide the foundation upon which governance policies, standard operating procedures, standards,

guidelines and supporting governance instruments will be developed and implemented across the Institute.

The Framework shall be guided by the following principles:

**1. Strategic alignment**

The IDGF shall support the strategic objectives of KIPRE and remain aligned with the Institute's mandate, Strategic Plan, applicable legal and regulatory requirements, and emerging national priorities relating to digital transformation, data governance and innovation.

**2. Institution-wide governance**

Data governance shall be implemented as an institution-wide responsibility. All directorates, divisions, programs, projects, sections and governance structures involved in the generation, collection, processing, storage, sharing and use of institutional data shall contribute to the implementation of the Framework in accordance with their respective mandates and responsibilities.

**3. Risk-based implementation**

Governance measures shall be proportionate to the nature, sensitivity, volume and intended use of institutional data. Higher-risk activities shall be subject to enhanced governance, oversight and compliance requirements.

**4. Life-cycle management**

Institutional data shall be governed throughout their entire life-cycle, including planning, collection, creation, storage, use, sharing, archival and secure disposal, while ensuring appropriate quality, integrity, security and accountability at each stage.

**5. Compliance by design**

Governance instruments developed under the Framework shall incorporate applicable legal, regulatory and institutional requirements from the outset, thereby promoting compliance by design rather than corrective implementation.

**6. Standardization and consistency**

Institutional governance processes shall be supported through standardized policies, SOPs, standards, guidelines, forms, templates, notices and registers to ensure consistency, transparency and efficiency across the Institute.

**7. Stewardship and accountability**

Clear governance roles, responsibilities and decision-making authority shall be assigned for institutional data assets. Data owners, custodians, stewards and users shall remain accountable for their respective governance responsibilities.

**8. Continual improvement**

The IDGF shall evolve progressively through regular monitoring, stakeholder engagement, periodic review, lessons learned and emerging best practices to ensure that governance ar-

rangements remain effective, relevant and responsive to changing institutional and technological environments.

## **2.9 Governance maturity pathway**

The implementation of the IDGF adopts a governance maturity approach that enables KIPRE to progressively strengthen its institutional data governance capabilities. Rather than attempting to establish the entire Framework simultaneously, implementation is structured into sequential phases that progressively build governance capacity, institutional ownership and organizational maturity.

Phase I establishes the foundational governance capability by operationalising the approved Data Management and Protection Policy and strengthening institutional data protection compliance. Phase II expands the governance framework through the development of additional governance policies and supporting governance instruments that collectively form the broader IDGF. Phase III focuses on institutionalizing the completed Framework through organization-wide implementation, capacity building, monitoring, evaluation and continual improvement.

This progressive implementation approach allows governance capabilities to mature in a coordinated and sustainable manner while ensuring that each phase provides the foundation for the successful implementation of the next.

## **2.10 Critical success factors**

Successful implementation of the IDGF will depend upon sustained institutional commitment and coordinated participation across all organizational levels. The following factors will be critical to the successful implementation of this Development Plan:

- Strong leadership commitment and management support.
- Clear governance structures with defined roles, responsibilities and accountability.
- Active participation of Directorates, Divisions and Sections in implementing approved governance instruments.
- Establishment of a functional network of Data Protection Champions to coordinate institutional compliance activities.
- Adequate financial, technical and human resources to support phased implementation.
- Continuous staff awareness, capacity building and change management.
- Regular monitoring, evaluation and continual improvement of institutional governance practices.

## **2.11 Phased development strategy**

The IDGF will be developed progressively through three implementation phases. Each phase builds upon the outputs of the preceding phase, allowing the Institute to establish governance capabilities incrementally while ensuring that immediate institutional priorities are addressed and available resources are utilized efficiently.

The phased approach recognizes that KIPRE has already established important governance foundations through the approval of the Data Management and Protection Policy, registration with the Office of the ODPC, establishment of the DPO Office, and the strategic commitment to develop an IDGF under the KIPRE Strategic Plan 2023–2027.

Consequently, the immediate priority is not the development of additional policies, but the operationalization of existing governance instruments before progressively expanding the governance framework into additional domains.

The three development phases are summarized below.

### **Phase I. Operationalization of the Data Management and Protection Policy**

This phase focuses on translating the approved Data Management and Protection Policy into operational practice. Emphasis will be placed on identifying the SOPs required under each policy article, developing priority procedures supporting the DPO Office, and producing the forms, templates, notices and institutional registers necessary to facilitate implementation.

### **Phase II. Expansion of the IDGF**

Following successful operationalization of the existing policy, the Framework will be expanded through the progressive development of additional governance policies covering priority institutional governance domains identified in PART I. Each policy will be accompanied by the operational and technical documentation required to support implementation.

### **Phase III. Institutionalization of the Framework**

The final phase focuses on integrating all governance instruments into a single institutional governance framework. Activities will include institutional roll-out, governance coordination, capacity building, monitoring and evaluation, continual improvement, and periodic review to ensure long-term sustainability of the Framework.

The outputs of each phase will collectively contribute towards the establishment of a comprehensive IDGF that supports effective governance of institutional data assets throughout their life-cycle while strengthening research excellence, institutional accountability, regulatory compliance and evidence-informed decision-making.

## **2.12 Phase I: Operationalizing the Data Management and Protection Policy**

This phase focuses on developing the governance instruments required to operationalize the approved Data Management and Protection Policy across the Institute.

### **2.12.1 Objectives**

The objectives of Phase I are to:

- a) Assess the Institute's current level of compliance with the DPA and the approved Data Management and Protection Policy.
- b) Develop the institutional Data Protection Compliance Implementation Strategy to guide the phased implementation of data protection governance across the Institute.
- c) Establish an institutional network of Data Protection Champions to coordinate implementation, promote awareness and support compliance activities within Directorates, Departments and Sections.
- d) Develop priority Standard Operating Procedures required for the effective operationalization of the Data Management and Protection Policy.
- e) Develop the supporting forms, templates, notices, agreements, registers and other controlled documents required to implement the approved Standard Operating Procedures.
- f) Develop a costed implementation work plan to support the phased implementation of approved governance instruments.
- g) Strengthen institutional compliance with the DPA, associated Regulations and the approved Data Management and Protection Policy.

### **2.12.2 Data protection compliance implementation strategy**

The successful operationalization of the Data Management and Protection Policy will require a coordinated institutional implementation strategy that extends beyond the Office of DPO. Effective implementation will depend on active participation by all Directorates, Departments and Sections responsible for generating, processing, storing or managing institutional data.

Accordingly, this phase (phase I) will establish an institutional data protection compliance implementation strategy that defines governance responsibilities, implementation priorities, reporting arrangements and institutional coordination mechanisms. The Office of DPO will provide overall technical leadership and regulatory oversight, while Heads of Directorates, Departments and Sections will be responsible for implementing approved governance instruments within their respective functional areas.

To facilitate institution-wide implementation, each Division will nominate a Data Protection Champion to serve as the focal person for coordinating compliance activities, supporting awareness programs, monitoring implementation, and liaising with the Office of the DPO. The Champions will provide an institutional coordination mechanism that promotes consistent implementation of the Policy while strengthening communication between operational units and the DPO Office.

The implementation strategy will also establish institutional reporting and monitoring arrangements to support progressive implementation of approved governance instruments, enabling KIPRE to monitor compliance, identify implementation challenges and continuously improve institutional data protection practices.

### 2.12.3 SOPs inventory

The successful implementation of the Data Management and Protection Policy requires documented SOPs that translate the Policy provisions into consistent institutional practice.

Accordingly, each substantive article of the Policy was systematically reviewed to identify the operational procedures necessary to support its implementation.

This review identified thirteen SOPs that collectively constitute the operational framework required to implement the current Data Management and Protection Policy. The proposed SOP inventory is presented in Table 3, together with the lead office responsible for each procedure, the corresponding policy provisions (articles) and the proposed development priority.

Recognizing that the development of all SOPs cannot be undertaken simultaneously, implementation will be phased according to institutional priorities, regulatory obligations and operational needs. SOPs assigned a **High** development priority constitute the initial implementation package to be developed during Phase I, while those assigned a **Medium** priority will be progressively developed as implementation of the Policy matures. Additional SOPs required to operationalize future governance policies will be developed during the subsequent expansion of the IDGF.



| Proposed SOP                                               | Lead office               | Policy article(s)  | Primary purpose                                                                                                                                                                    | Dev. priority |
|------------------------------------------------------------|---------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| SOP-001: Data Protection Compliance Review                 | DPO Office                | Articles 2–4       | Establish procedures for reviewing research, surveillance, consultancy, procurement, ICT systems and other institutional activities involving personal data before implementation. | High          |
| SOP-002: Data Subject Rights Management                    | DPO Office                | Articles 3–4       | Operationalize requests relating to access, rectification, restriction, objection, portability and erasure of personal data.                                                       | High          |
| SOP-003: Data Protection Impact Assessment (DPIA)          | DPO Office                | Articles 2, 3 & 10 | Establish procedures for conducting Data Protection Impact Assessments for high-risk processing activities.                                                                        | High          |
| SOP-004: Personal Data Management                          | DPO Office                | Article 4          | Standardize the lawful collection, processing, storage, access and management of personal data throughout its life-cycle.                                                          | High          |
| SOP-005: Research Data Management (RDM)                    | RPDD / DSAS               | Article 4          | Establish procedures for research data stewardship, investigator responsibilities, metadata management and research data life-cycle management.                                    | High          |
| SOP-006: Data Sharing and Disclosure                       | DPO Office / Legal Office | Article 5          | Govern the internal, external and international sharing of institutional data while ensuring compliance with legal and contractual obligations.                                    | High          |
| SOP-007: Information Security and Data Protection Controls | ICT Division              | Article 6          | Define minimum technical, organizational and physical safeguards for protecting institutional information assets and personal data.                                                | Medium        |

| Proposed SOP                                                 | Lead office                 | Policy article(s) | Primary purpose                                                                                                                        | Dev. priority |
|--------------------------------------------------------------|-----------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|---------------|
| SOP-008: Personal Data Breach Management                     | DPO Office                  | Article 7         | Establish procedures for breach reporting, investigation, notification, containment and corrective actions.                            | High          |
| SOP-009: Data Retention, Archiving and Secure Disposal       | DPO Office / Registry / ICT | Article 8         | Standardize retention schedules, archival procedures and secure disposal of institutional data and records.                            | Medium        |
| SOP-010: Data Protection Complaints and Dispute Resolution   | DPO Office                  | Article 9         | Establish procedures for receiving, investigating and resolving complaints relating to personal data processing.                       | Medium        |
| SOP-011: Data Protection Monitoring, Audit and Compliance    | DPO Office                  | Articles 10–12    | Operationalize compliance monitoring, internal audits, reporting and continual improvement of institutional data protection practices. | Medium        |
| SOP-012: Third-Party Data Processing and Contract Management | Legal and DPO Offices       | Articles 3 & 5    | Govern the engagement, monitoring and oversight of third-party processors and contractual data protection obligations.                 | Medium        |
| SOP-013: Data Protection Awareness and Training              | CBPGMD & DPO Office         | Articles 2 & 6    | Standardize institutional awareness, induction and periodic data protection training for staff, students and collaborators.            | Medium        |

Table 3: Proposed SOPs required to operationalize the Data Management and Protection Policy.

### 2.12.4 Priority SOPs for the DPO Office

Although the operationalization of the Data Management and Protection Policy requires the progressive development of all identified SOPs, the initial implementation effort will focus on those that directly support the statutory and operational responsibilities of the DPO office. These procedures establish the institutional mechanisms required to implement the DPA and provide the operational foundation upon which the broader IDGF will be progressively developed.

The priority SOPs have been selected based on three considerations. First, they address the statutory obligations imposed on the Institute as a Data Controller and Data Processor under the DPA. Second, they establish the core institutional processes required for lawful processing of personal data across research, surveillance, consultancy and administrative activities. Third, they provide the governance infrastructure upon which the remaining institutional SOPs can be implemented in a coordinated and consistent manner.

The priority SOPs proposed for immediate development are presented in Table 4.

| Priority SOP                             | Primary justification                                                                                                                                                             |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Protection Compliance Review        | Provides the institutional entry point for reviewing research, consultancy, surveillance and other activities involving personal data before implementation.                      |
| Data Subject Rights Management           | Establishes procedures for responding to requests relating to access, rectification, restriction, objection, portability and erasure of personal data in accordance with the DPA. |
| Data Protection Impact Assessment (DPIA) | Provides a structured methodology for assessing privacy risks associated with high-risk processing activities before implementation.                                              |
| Personal Data Management                 | Standardizes the collection, processing, storage, use and management of personal data throughout its life-cycle.                                                                  |
| Data Sharing and Disclosure              | Establishes institutional procedures governing the lawful sharing of personal data within KIPRE and with external collaborators.                                                  |
| Personal Data Breach Management          | Provides procedures for breach detection, containment, investigation, statutory notification and corrective action.                                                               |

Table 4: Priority SOPs for the DPO Office.

Collectively, these six (6) SOPs constitute the minimum operational package required to establish the DPO Office as the institutional focal point for privacy governance and regulatory compliance. Their implementation will provide the immediate operational capability required to support institutional compliance while creating the governance foundation upon which the remaining SOPs identified in Table 3 will be progressively developed.

### 2.12.5 Supporting implementation documents

The implementation of the priority Standard Operating Procedures requires a suite of controlled implementation documents that support consistent execution, documentation, communication and institutional accountability. These documents provide the operational tools necessary for implementing approved procedures, recording institutional decisions, demonstrating compliance and maintaining appropriate governance records.

The implementation documents developed during Phase I will primarily support the operational functions of the Office of the Data Protection Officer. As additional governance policies are developed under subsequent phases of the IDGF, corresponding implementation documents will be developed to support those governance instruments.

| <b>Priority SOP</b>                      | <b>Proposed implementation documents</b>                                                                                                                                                                                                                                                | <b>Responsible office</b>                |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| Data Protection Compliance Review        | <ul style="list-style-type: none"> <li>• Data Protection Compliance Review Form</li> <li>• Compliance Review Checklist</li> <li>• Compliance Review Register</li> <li>• Compliance Decision Letter</li> </ul>                                                                           | Office of the DPO                        |
| Data Protection Impact Assessment (DPIA) | <ul style="list-style-type: none"> <li>• DPIA Screening Form</li> <li>• DPIA Assessment Template</li> <li>• DPIA Report Template</li> <li>• DPIA Register</li> </ul>                                                                                                                    | Office of the DPO                        |
| Data Subject Rights Management           | <ul style="list-style-type: none"> <li>• Data Subject Request Form</li> <li>• Identity Verification Form</li> <li>• Response Letter Templates</li> <li>• Data Subject Rights Register</li> </ul>                                                                                        | Office of the DPO                        |
| Personal Data Management                 | <ul style="list-style-type: none"> <li>• Privacy Notice Templates</li> <li>• Consent Form Templates</li> <li>• Withdrawal of Consent Form</li> <li>• Privacy Statements</li> </ul>                                                                                                      | Office of the DPO / Research Directorate |
| Research Data Handling and Protection    | <ul style="list-style-type: none"> <li>• Research Data Management Plan Template</li> <li>• Data Inventory Register</li> <li>• Metadata Template</li> <li>• Data Custody Form</li> </ul>                                                                                                 | Research Directorate / DSAS              |
| Data Sharing and Disclosure              | <ul style="list-style-type: none"> <li>• Data Sharing Request Form</li> <li>• Data Sharing Agreement (DSA)</li> <li>• Data Processing Agreement (DPA)</li> <li>• Non-Disclosure Agreement (NDA)</li> <li>• Data Transfer Agreement (DTA)</li> <li>• Data Disclosure Register</li> </ul> | Legal Office / Office of the DPO         |
| Personal Data Breach Management          | <ul style="list-style-type: none"> <li>• Personal Data Breach Report Form</li> <li>• Internal Incident Report</li> <li>• ODPC Breach Notification Template</li> <li>• Personal Data Breach Register</li> </ul>                                                                          | Office of the DPO                        |

| Priority SOP                                      | Proposed implementation documents                                                                                                                                                          | Responsible office      |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Information Security and Data Protection Controls | <ul style="list-style-type: none"> <li>• User Access Request Form</li> <li>• Access Control Register</li> <li>• ICT Asset Register</li> <li>• Secure Media Disposal Certificate</li> </ul> | ICT / Office of the DPO |

Table 5: Proposed supporting implementation documents for the Phase I priority Standard Operating Procedures.

### 2.12.6 Expected outputs

Implementation of Phase I is expected to produce the following outputs:

- a) A fully operational implementation framework supporting the Data Management and Protection Policy.
- b) Priority SOPs for the Office of the DPO developed and approved.
- c) Standardized forms, templates, notices and institutional registers supporting implementation of the priority SOPs.
- d) Standardized institutional procedures for compliance review, personal data management, data sharing, data subject rights management, data breach management and Data Protection Impact Assessments.
- e) Improved institutional compliance with the DPA, the Data Protection Regulations and the KIPRE Data Management and Protection Policy.
- f) A governance foundation for the progressive expansion of the IDGF.

## 2.13 Phase II: Expansion of the governance framework

Having established the operational foundation for the Data Management and Protection Policy under Phase I, the next stage focuses on the development of the broader IDGF. This phase aims to expand institutional data governance beyond data protection by developing the additional governance instruments required to support the effective management, stewardship and utilization of institutional data assets.

Given the multidisciplinary and technical nature of institutional data governance, this phase will involve the engagement of a technical consultant to work collaboratively with KIPRE in designing and developing the remaining governance framework. The consultant will work closely with the Office of DPO, the DSAS, ICT, the RPDD and other relevant stakeholders to ensure that the Framework aligns with national legislation, international best practices and the strategic objectives of the Institute.

### 2.13.1 Purpose of Phase II

The purpose of Phase II is to progressively develop the remaining components of the IDGF by establishing additional governance policies together with their supporting implementation instruments. Collectively, these governance instruments will provide the institutional architecture required for governing research data, institutional information assets, data quality, AI, information security and other strategic data resources across their life-cycle.

### 2.13.2 Governance policies to be developed

The proposed governance policies to be developed under Phase II are presented in Table 6 and as also outlined in Table 2 of Part I.

| Governance policy                    | Primary focus                                                                                                                              |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Research Data Management Policy      | Governance of research data throughout its life-cycle, including ownership, stewardship, preservation, sharing and reuse.                  |
| Data Quality Management Policy       | Institutional standards for ensuring the accuracy, completeness, consistency, timeliness and integrity of data.                            |
| Information Governance Policy        | Enterprise governance of institutional information assets, including accountability, classification and information life-cycle management. |
| AI Governance Policy                 | Responsible development, acquisition, deployment and oversight of Artificial Intelligence and Machine Learning systems.                    |
| Open Science and Data Sharing Policy | Institutional governance of scientific data sharing, open science practices and collaborative research.                                    |

Table 6: Proposed governance policies for development under Phase II.

### 2.13.3 Supporting governance instruments

Each governance policy developed under Phase II will be accompanied by a suite of implementation documents necessary to support consistent institutional application. These governance instruments will include:

- Standard Operating Procedures (SOPs);
- Standards;
- Guidelines;
- Forms;
- Templates;
- Notices;
- Checklists; and
- Registers.

The specific implementation instruments required for each governance policy will be identified during the policy development process and incorporated into the IDGF.

### 2.13.4 Governance structures and stewardship

Successful implementation of the IDGF requires clearly defined governance responsibilities across the Institute. Table 7 presents the proposed governance structure for institutional data stewardship.

| Governance function | Lead responsibility                                                                                          |
|---------------------|--------------------------------------------------------------------------------------------------------------|
| Office of the DPO   | Privacy governance, regulatory compliance and oversight of personal data protection.                         |
| DSAS                | Institutional data governance, data stewardship, data quality and analytics governance.                      |
| ICT                 | Information security, technical infrastructure, access management and cybersecurity controls.                |
| RPDD                | Governance of research data, scientific integrity and research compliance.                                   |
| Directorate Heads   | Implementation of governance requirements within their respective directorates.                              |
| Data Stewards       | Operational oversight of institutional datasets, metadata and data quality within assigned functional areas. |
| Data Custodians     | Management, storage and protection of institutional data assets under delegated authority.                   |

Table 7: Proposed governance structure and stewardship responsibilities under the IDGF.

## **2.14 Phase III: Institutionalization of the IDGF**

Following the development of the governance policies and supporting implementation instruments under Phase II, the final stage focuses on institutionalizing the completed IDGF across KIPRE. The objective of this phase is to integrate data governance into routine institutional operations, ensuring that governance principles become part of organizational culture, decision-making processes and day-to-day business activities.

Institutionalization extends beyond the publication of governance documents. It requires organizational commitment, clearly defined governance responsibilities, continuous capacity building, regular monitoring of compliance, periodic review of governance instruments and continual improvement of the Framework in response to emerging institutional, technological and regulatory requirements.

### **2.14.1 Framework integration**

The completed IDGF will be integrated into existing institutional governance, research, administrative and ICT processes. Governance requirements will be embedded within project approval processes, research management, information systems, procurement, partnerships and institutional decision-making to ensure that data governance becomes an integral component of organizational operations.

### **2.14.2 Capacity building**

Successful institutionalization of the Framework will require continuous awareness creation and competency development across all directorates and divisions. Training programs will be implemented for staff, researchers, students and collaborators to strengthen understanding of institutional governance policies, operational procedures and individual responsibilities relating to data governance and data protection.

### **2.14.3 Monitoring and evaluation**

Implementation of the IDGF will be monitored through periodic compliance assessments, internal audits, management reviews and performance reporting. Monitoring activities will evaluate adherence to governance policies, effectiveness of operational procedures, implementation of corrective actions and overall maturity of institutional data governance.

Appropriate governance indicators and performance measures will be developed to assess implementation progress and identify opportunities for continual improvement.

### **2.14.4 Continuous improvement**

The IDGF will remain a living governance system that evolves in response to changes in legislation, institutional priorities, technological developments and emerging best practices. Governance policies, SOPs, standards and supporting documentation will therefore be reviewed periodically to ensure their continued relevance, effectiveness and alignment with the strategic objectives of the Institute.

Lessons learned during implementation, audit findings, stakeholder feedback and technological advancements will inform future revisions of the Framework to ensure that KIPRE maintains a mature,



adaptive and sustainable institutional data governance program.

### 3 Overall implementation road-map

The implementation of the IDGF will be undertaken progressively through three complementary phases. Each phase builds upon the achievements of the preceding phase to ensure that governance capabilities mature in a structured, coordinated and sustainable manner (see Table 8 below).

| Phase     | Major activities                                                                                                                              | Lead responsibility                              | Expected outcome                                                                          |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|-------------------------------------------------------------------------------------------|
| Phase I   | Operationalize the Data Management and Protection Policy through the development of priority SOPs and supporting implementation documents.    | Office of the DPO                                | Functional institutional data protection governance and operational compliance framework. |
| Phase II  | Develop the remaining governance policies, standards, guidelines, SOPs and supporting governance instruments through a technical consultancy. | Technical Consultant in collaboration with KIPRE | Comprehensive IDGF established.                                                           |
| Phase III | Institutionalize the completed Framework through implementation, capacity building, monitoring, evaluation and continual improvement.         | KIPRE Management                                 | Sustainable enterprise-wide institutional data governance program.                        |

Table 8: Overall implementation road-map for the IDGF.

### 4 Conclusion

The development of the IDGF a significant step towards strengthening the governance, management and utilization of data across KIPRE. By adopting a phased implementation approach, the Institute will progressively establish the governance policies, operational procedures, technical standards and supporting governance instruments required to achieve a mature and sustainable data governance program.

The successful implementation of this Development Plan will not only operationalize the existing Data Management and Protection Policy but will also provide the governance architecture necessary to support research excellence, regulatory compliance, institutional accountability and responsible innovation. Through continual implementation, monitoring and improvement, the IDGF will evolve into an integrated enterprise-wide governance framework that promotes the effective stewardship of institutional data assets and supports the strategic objectives of the Institute.

## 5 References

- [1] Constitution of Kenya. (2010). Government of Kenya.
- [2] Data Protection Act, No. 24 of 2019. Government of Kenya.
- [3] Data Protection (General) Regulations, 2021. Government of Kenya.
- [4] Computer Misuse and Cybercrimes Act, No. 5 of 2018. Government of Kenya.
- [5] Access to Information Act, No. 31 of 2016. Government of Kenya.
- [6] Kenya Information and Communications Act (Cap. 411A). Government of Kenya.
- [7] Public Archives and Documentation Service Act (Cap. 19). Government of Kenya.
- [8] Ministry of Information, Communications and the Digital Economy. (2022). Kenya National Digital Master Plan 2022–2032.
- [9] Ministry of Information, Communications and the Digital Economy. (2025). Kenya National Artificial Intelligence Strategy 2025–2030.
- [10] DAMA International. (2017). *The DAMA Guide to the Data Management Body of Knowledge (DAMA-DMBOK2)* (2nd ed.). Technics Publications.
- [11] Organisation for Economic Co-operation and Development (OECD). (2023). *Data Governance in the Public Sector*. OECD Publishing.
- [12] UNESCO. (2021). *Recommendation on the Ethics of Artificial Intelligence*. United Nations Educational, Scientific and Cultural Organization.
- [13] International Organization for Standardization (ISO). (2017). *ISO/IEC 38505-1: Information technology — Governance of IT — Governance of data*.
- [14] International Organization for Standardization (ISO). (2022). *ISO/IEC 27001: Information security, cybersecurity and privacy protection — Information security management systems — Requirements*.
- [15] International Organization for Standardization (ISO). (2019). *ISO/IEC 27701: Privacy Information Management Systems*.
- [16] Wilkinson, M. D., Dumontier, M., Aalbersberg, I. J., et al. (2016). The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data*, 3, 160018.
- [17] UNESCO. (2021). *UNESCO Recommendation on Open Science*. United Nations Educational, Scientific and Cultural Organization.
- [18] Kenya Institute of Primate Research (KIPRE). (2023). *Strategic Plan 2023–2027*.
- [19] Kenya Institute of Primate Research (KIPRE). (2026). *Data Management and Protection Policy*.
- [20] Kenya Institute of Primate Research (KIPRE). (2026). *Institutional Data Management Needs Assessment Report*.
- [21] Office of the Data Protection Commissioner (ODPC). (2026). *Certificate of Registration as Data Controller and Data Processor*.